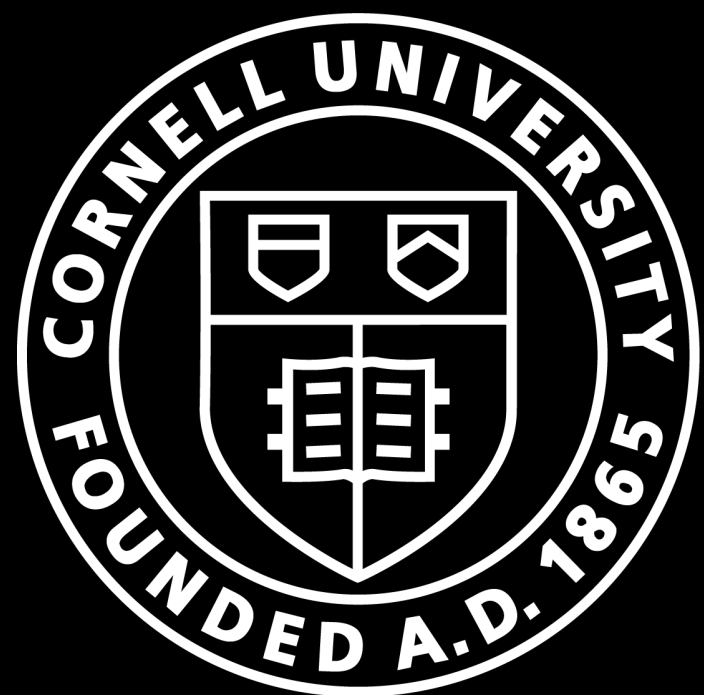




Lecture 1a: The Beginning. Course policies, procedures, and expectations. Motivating the study of quantitative finance.

CHEME 5660 Cornell University

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Disclaimer and Risks

Training only. This content is for training and informational purposes only. The teaching team does not make, give, or endorse any offer or solicitation to buy or sell securities or derivative products or provide investment or trading advice or strategy.

Risk. Trading involves risk and is generally not recommended for those with limited resources, experience, or a low-risk tolerance. Past performance does not guarantee future success.

Responsibility. You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.



First two weeks: In this course, we start by introducing financial basics, personal and corporate finance tools, and topics.

Prof. Woltornist aw499@cornell.edu



The remaining 14 x weeks: we focus on quantitative finance, and the modeling, and analysis of wealth creation and management using debt, equity, and derivatives

Prof. Varner jdv27@cornell.edu



Teaching Assistant: Aaron Wheeler, Ph.D. student in the Varnerlab, is an emerging expert in market microstructure and market simulation

Aaron Wheeler aw843@cornell.edu



Course information: Policies, expectations, and materials for the course are hosted online and can be found at: <http://bit.ly/3DSF0of>

Huge data scale.

- ▶ During peak trading times, approximately *36 GB of data is produced per minute* from equity trading in the United States. Peak data generation is around 14 TB per day x 252 trading days per year (3.5 petabytes per year for equity only, not including options).
 - The Library of Congress holds approximately 20 petabytes of data
 - If you took a petabyte's worth of 1GB flash drives and lined them up end to end, they would stretch over 92 football fields.
 - The average 4k movie is 100GB of data. Thus, 1 x petabyte of storage could hold 11,000 4k movies. With an average run time of 2 hours, it would take you 2.5 years of nonstop binge-watching to get through a petabyte's worth of 4k movies.
- ▶ We can access price and volume data (in real-time) at the resolution of a second. In some (extreme) cases, we can access data at nano-second frequencies.
- ▶ *What can we do with that data?*

Huge financial scale.



The global biopharma market: The annual global biopharmaceuticals market is valued at USD 478.20 billion in 2023. It is expected to reach USD 704.91 billion in 2028, recording a CAGR of 8.07% over the forecast period (2023-2028).

<https://www.mordorintelligence.com/industry-reports/global-biopharmaceuticals-market-industry>

Huge financial scale.



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The options market in the United States: The *average daily notional value of traded single-stock options rose to more than \$479 billion in 2023* compared with about \$405 billion for stocks (2021), according to Cboe Global Markets data (2021 - 2023)

<https://qz.com/2092197/options-trading-is-poised-to-overtake-the->

[Computation: Average Market Value x L](#)

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[Computation: Average Market Value x L](#)

In a single day, the options market in the United States trades, on average, at least the annual global biopharmaceutical market.

A blue smartphone is standing upright in a field of green grass. The phone is positioned vertically, with its back to the camera, showing the triple-camera system. The background is a soft-focus landscape with green foliage and a bright sun in the upper right corner, creating a lens flare effect. A semi-transparent dark blue rectangular box is overlaid on the center of the image, containing white text.

The barrier to entry to markets and advanced financial products has vanished.

You need a bank account and a smartphone.

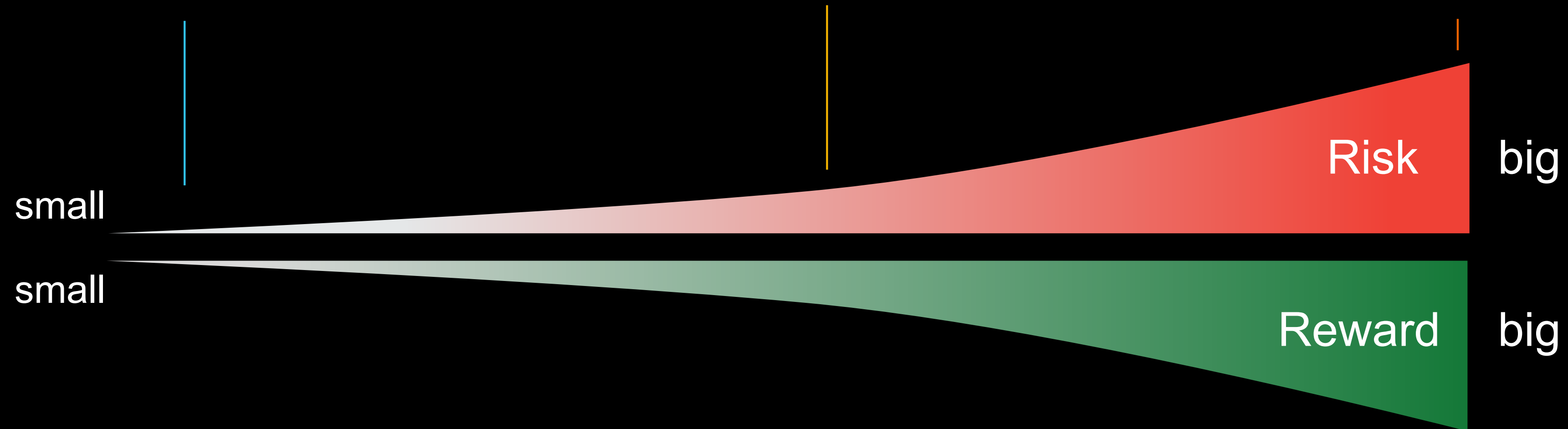
Fixed Income Debt Securities (US Treasury Bills, Notes, and Bonds)



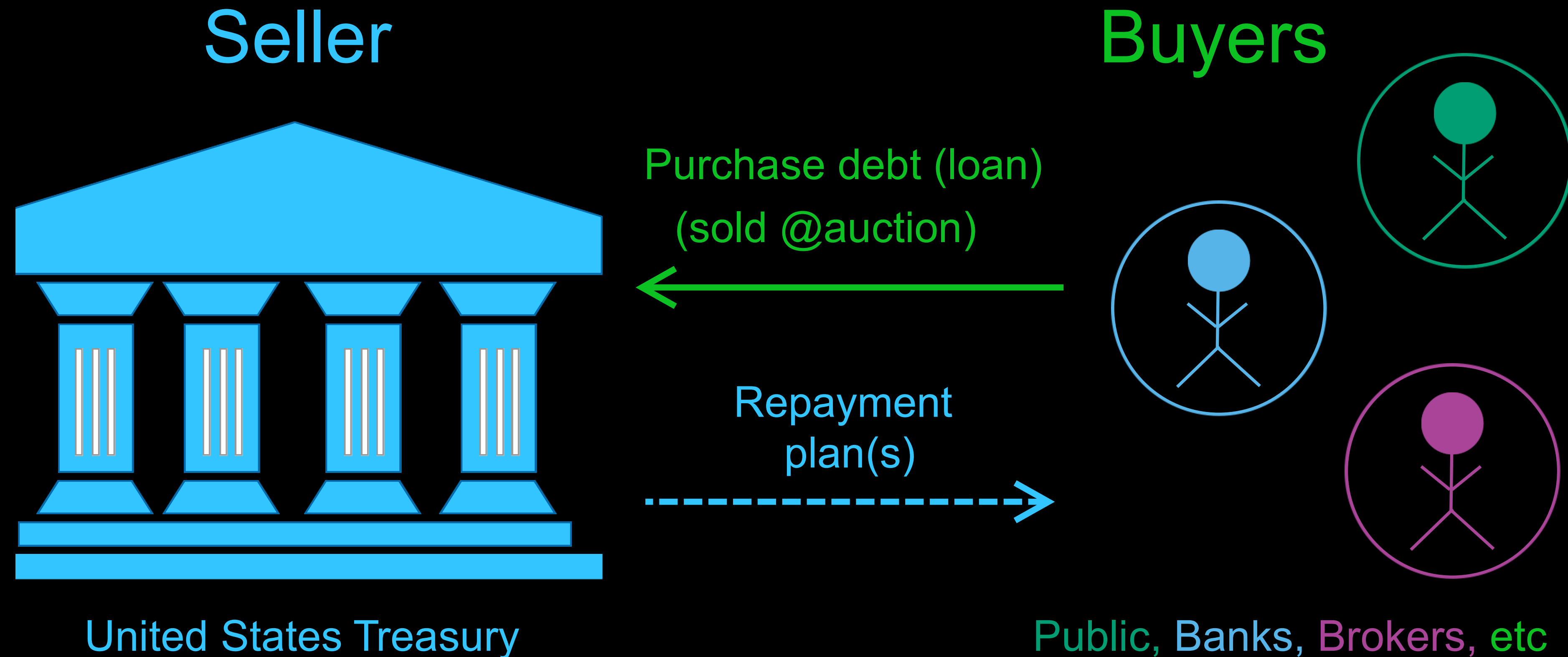
Equity Securities (Individual and Baskets of Stocks)



Derivative Securities (Individual and Collections of Equity Options Contracts)



Relationship between the United States Treasury and Debt holders. The U.S. Government uses U.S. Treasury debt securities to borrow money from bondholders with fixed repayment plans.



U.S. Treasury debt securities are one of the safest possible investments (*risk-free?*). U.S. Treasury debt securities are backed by the full faith and credit of the United States government; no defaults in recent memory.

<https://www.treasurydirect.gov/>

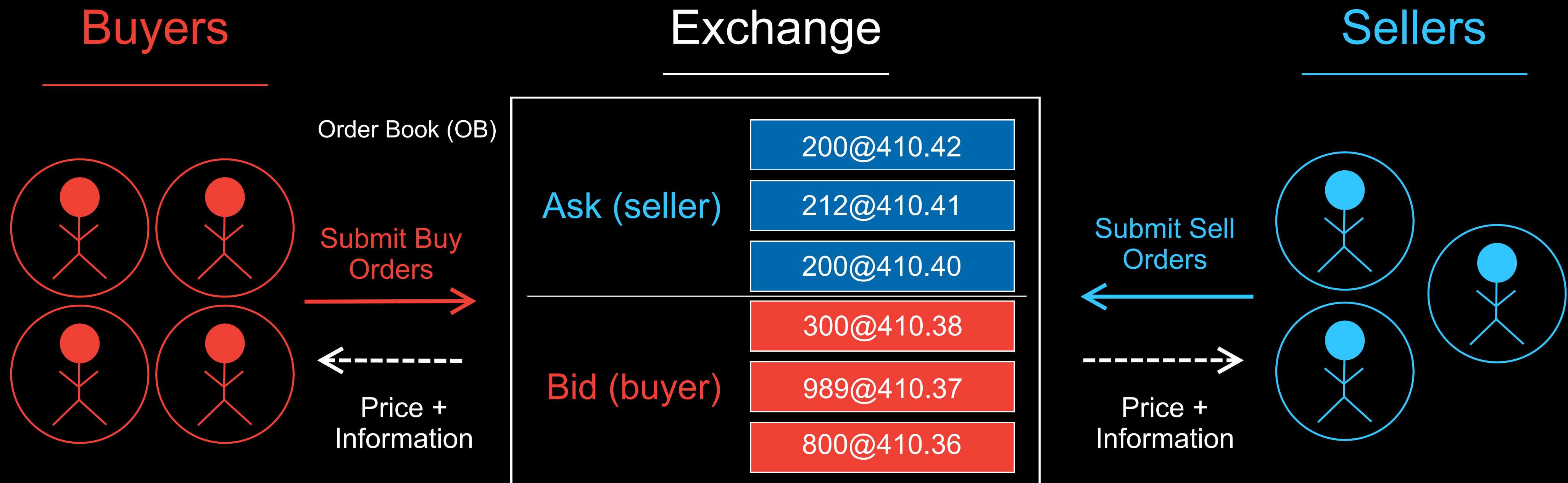
Understanding **Interest Rate Risk**. The factors behind Silicon Valley Bank's collapse, PBS News Hour Mar 28, 2023, 6:45 PM EDT



Understanding **Interest Rate Risk**. The factors behind Silicon Valley Bank's collapse, PBS News Hour Mar 28, 2023, 6:45 PM EDT





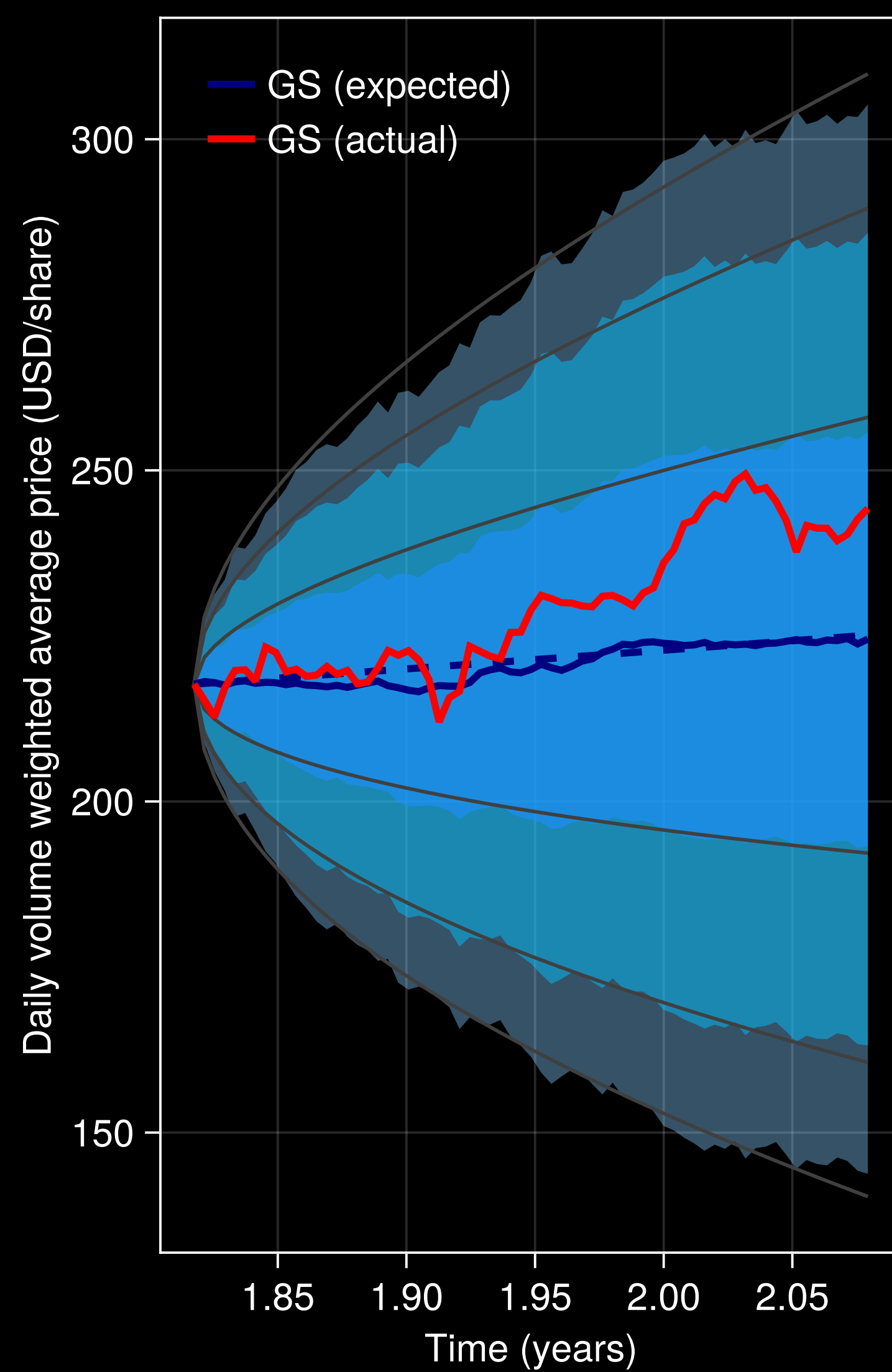
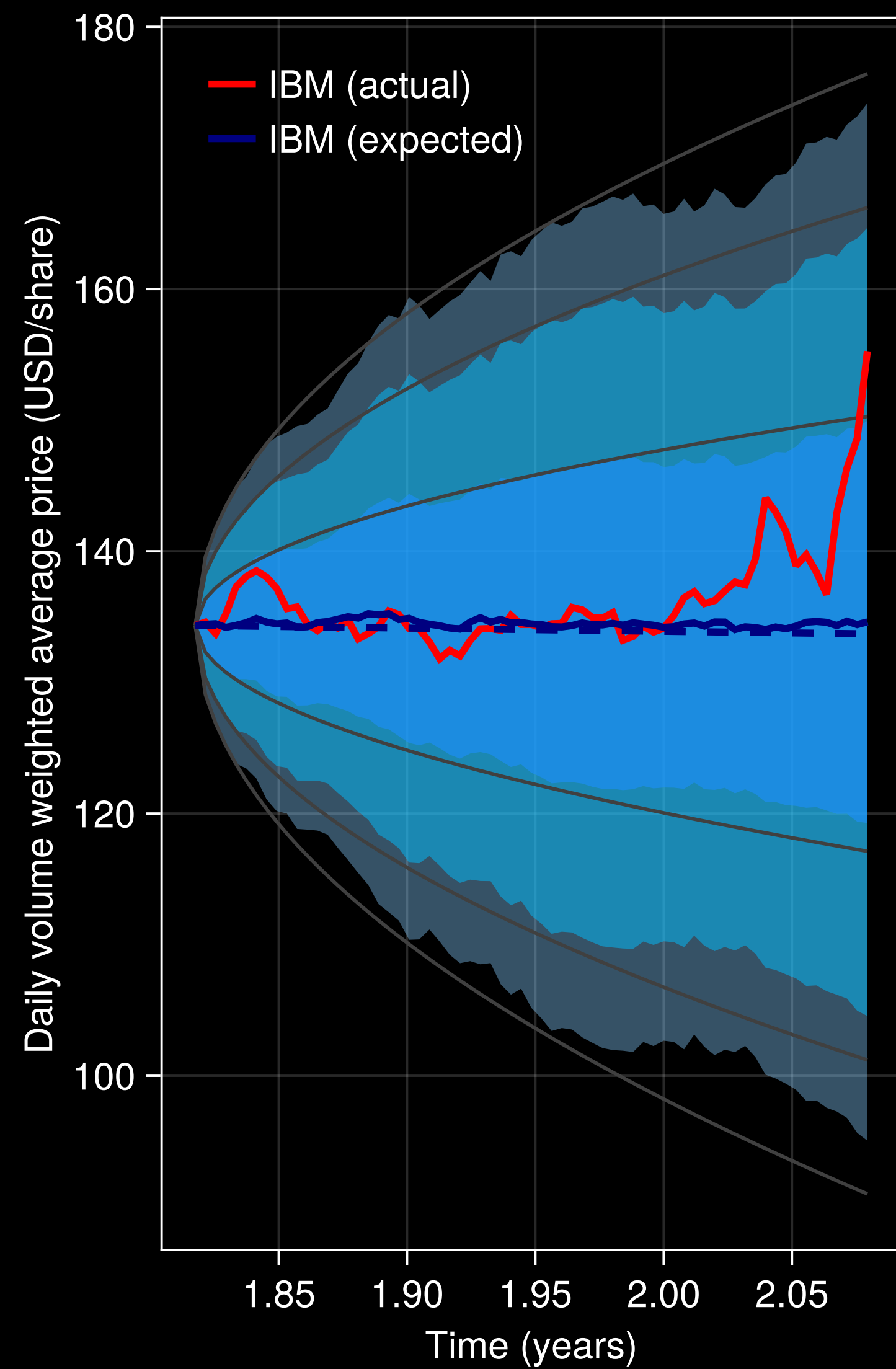


What Is a Market Maker?

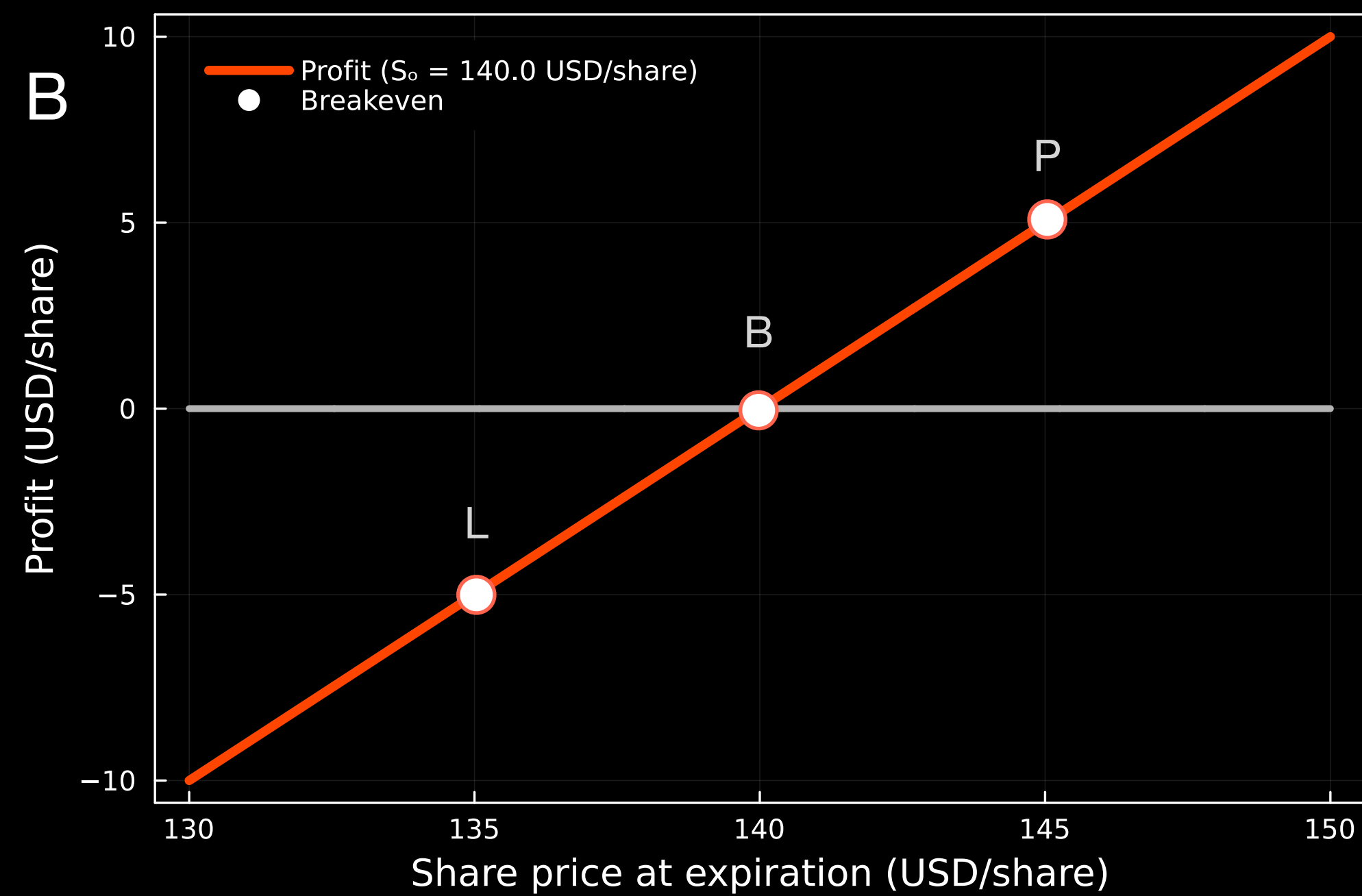
What Is a Market Maker?

DEMO Level 2 order book data: AMD

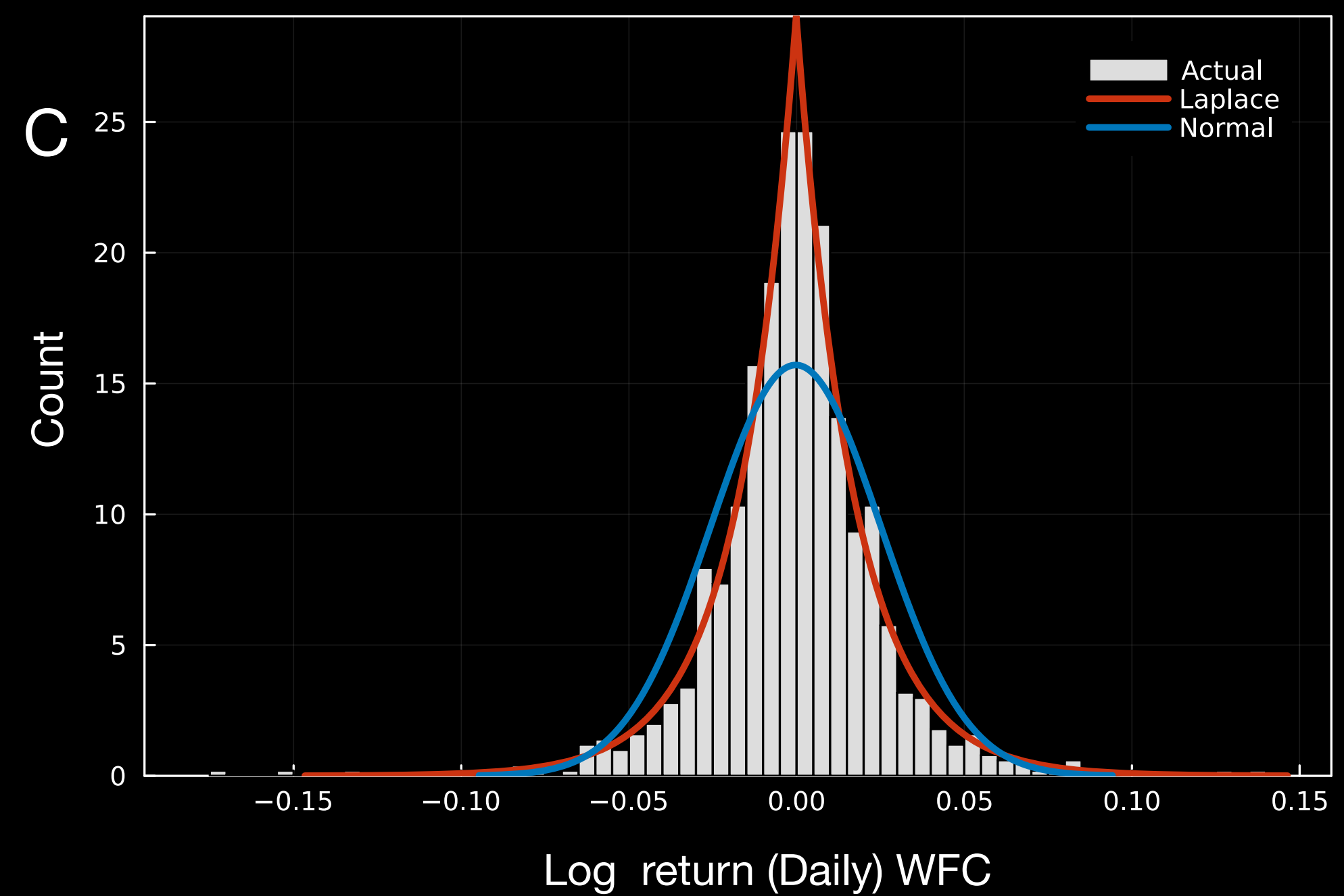
A



B



C

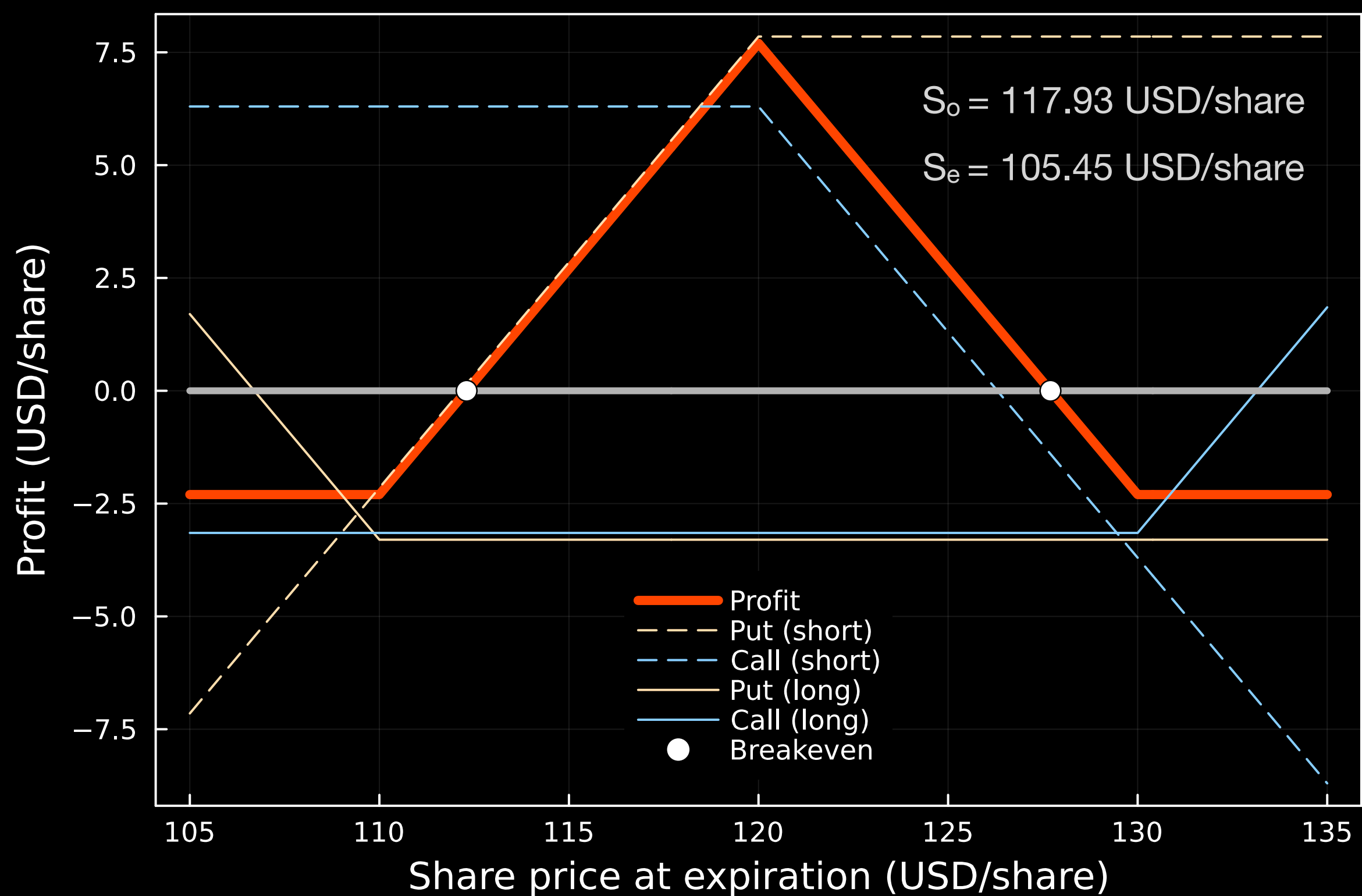


Derivatives. Equity derivatives (options) are temporary structured agreements between a buyer and a seller to transact (buy or sell) an underlying asset (shares of stock) at a specified price

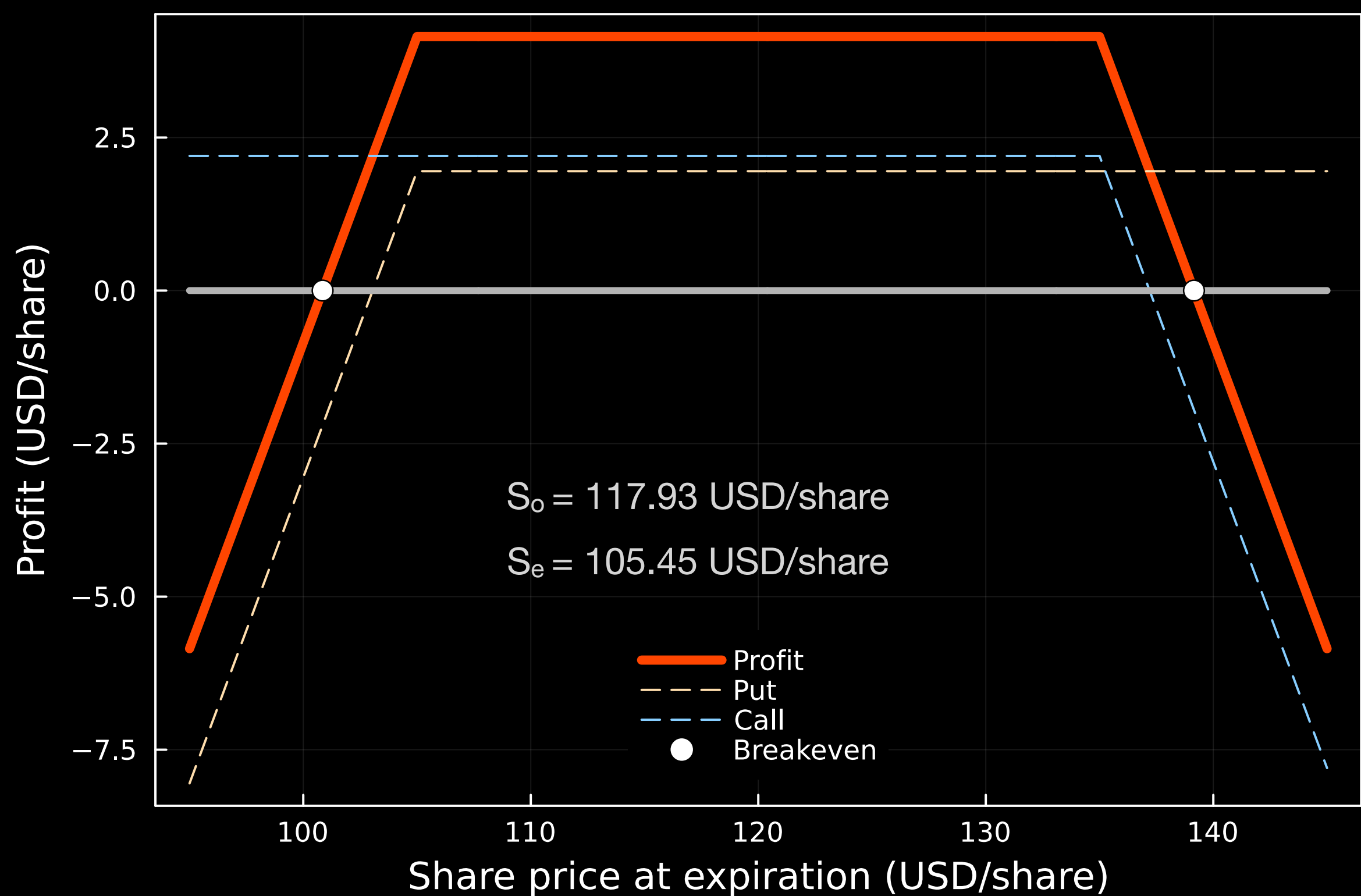


Equity derivatives (options) of different types (call/put) and orientations (short/long) can be combined to produce trade structures with complex payoff/profit diagrams.

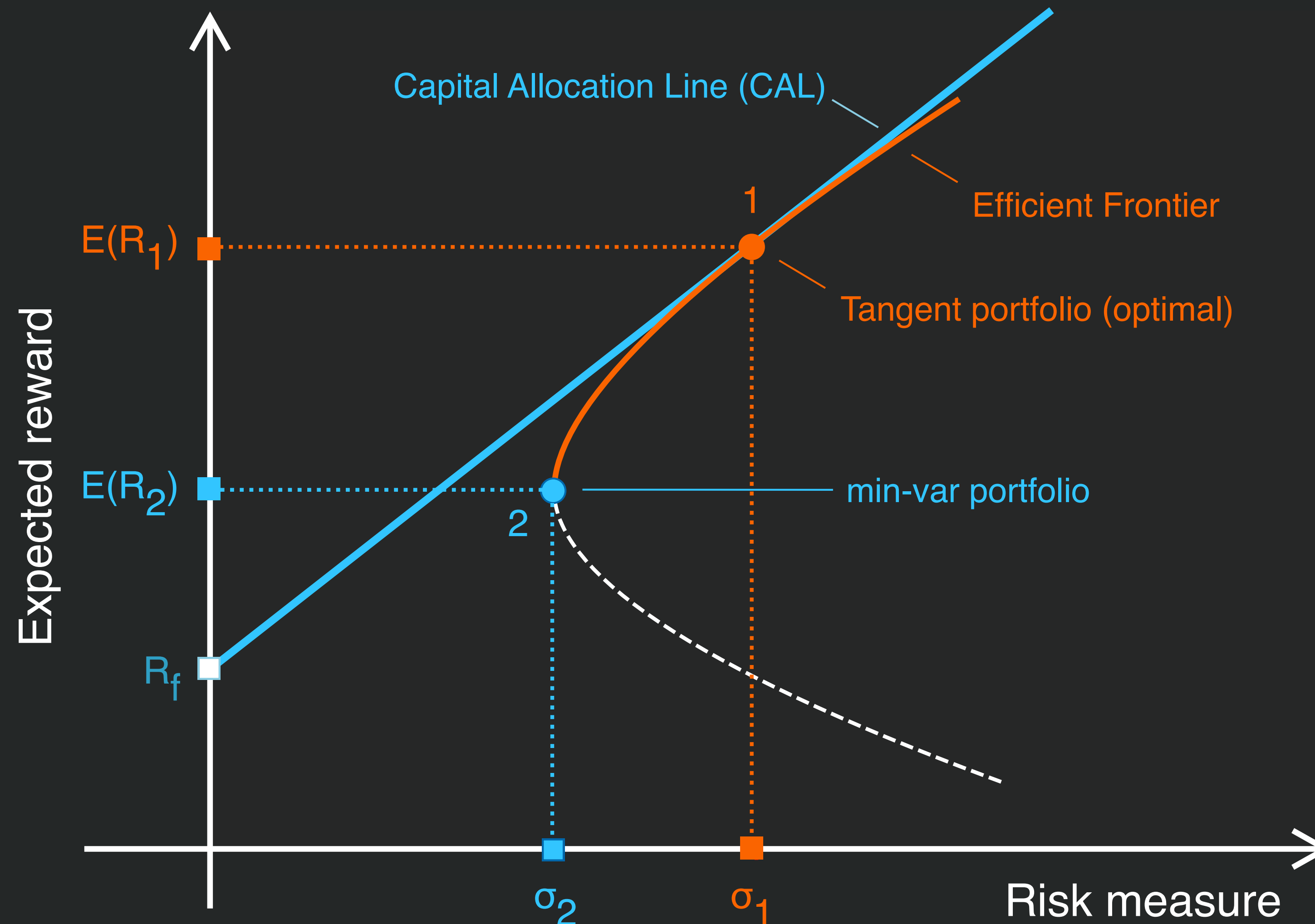
Short Iron butterfly DTE = 31 days
(defined risk-neutral trade)



Short Strangle DTE = 31 days
(unlimited risk neutral trade)



Optimally balance risk and reward in an asset portfolio. Find the collection of assets that minimizes the risk for a desired level of return



Fixed Income Debt Securities (US Treasury Bills, Notes, and Bonds)



Interest rate data since 1954 (including daily data) and US Treasury Auction Data

Data provider: Federal Reserve
+ WSJ + Various

Equity Securities (Individual and Baskets of Stocks)



OHLC equity daily data for every component of the SP500 for the past 5-years + live 2023 data

Data provider: polygon.io

Derivative Securities (Individual and Collections of Equity Options Contracts)



Option chain data (weekly and monthly) for every member of the SP500 that offers options

Today?